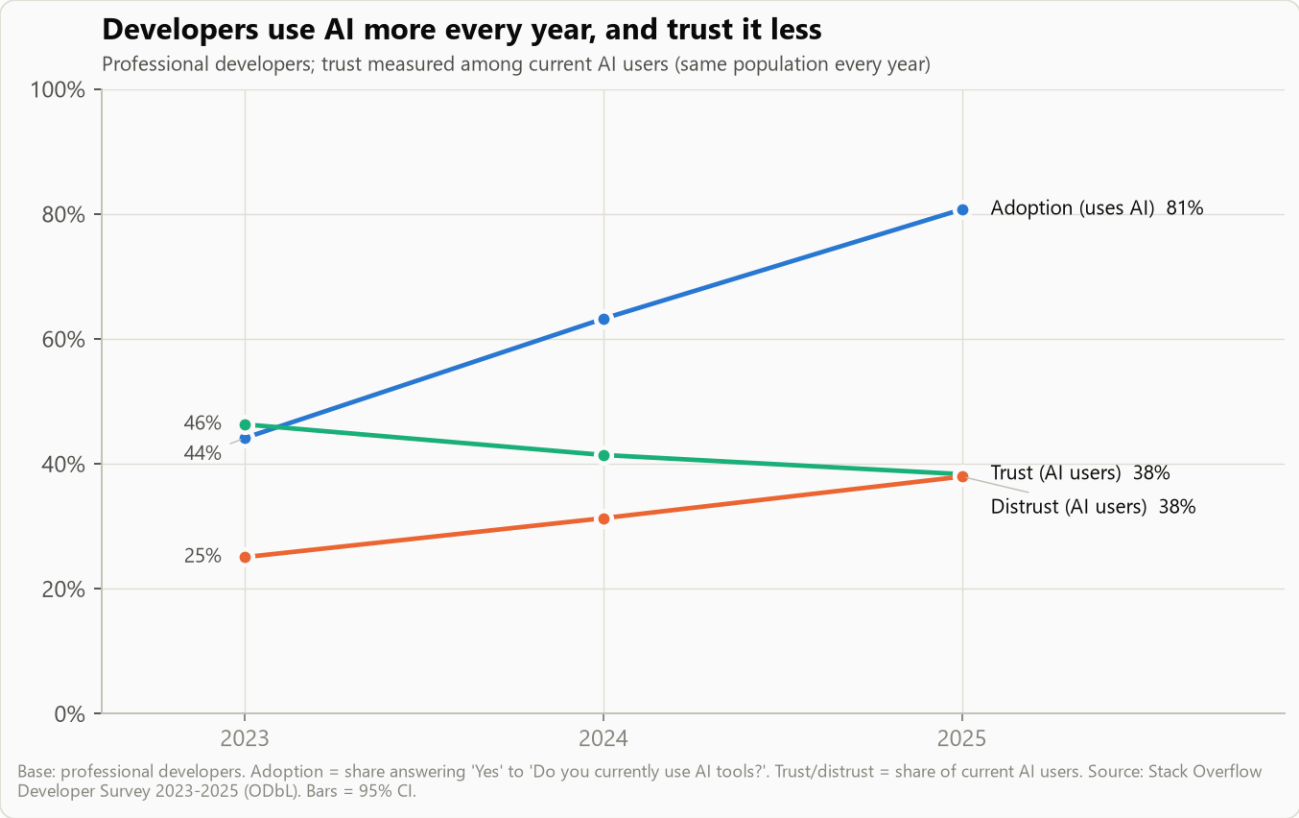
Developers use AI more every year, and trust it less

Insight memo · Stack Overflow Developer Survey 2023–2025 · For developer-tools product teams and engineering leaders

> **Bottom line.** Among professional developers, AI adoption nearly doubled in two years (44% → 81%). Over the same period, distrust of AI output among people who *use* AI rose from 25% to 38%. By 2025, users are as likely to distrust AI accuracy as to trust it.
>
> Distrust is highest among **experienced developers**, **occasional users** and people for whom **fixing AI output is slow**. The widely quoted "almost right, but not quite" frustration is real, but it is nearly universal (76% of professional AI users), so on its own it does **not** separate trusters from distrusters. **The cost of debugging AI output is what tracks distrust.**



Key findings

1. The trust decline is real and not just "a different crowd". Among professional AI users, trust fell from 46% (2023) to 41% (2024) to 38% (2025), and distrust rose from 25% to 31% to 38%.

- **Holding the respondent mix constant:** with experience, role, company size, country and age controlled, the odds of distrust were **1.87× higher in 2025 than in 2023** (+13 percentage points).
- **After reweighting** 2024 and 2025 to look like the 2023 sample, distrust still rises from 25% to 39%.
- **Caution on the official numbers:** Stack Overflow's own tables show trust as flat from 2023 to 2024 (42% → 43%) and then falling to 33% in 2025. Those tables use a different population each

year: users + planners in 2023, users only in 2024, everyone including non-users in 2025. Comparing like with like (current AI users) shows a steady decline in every year.

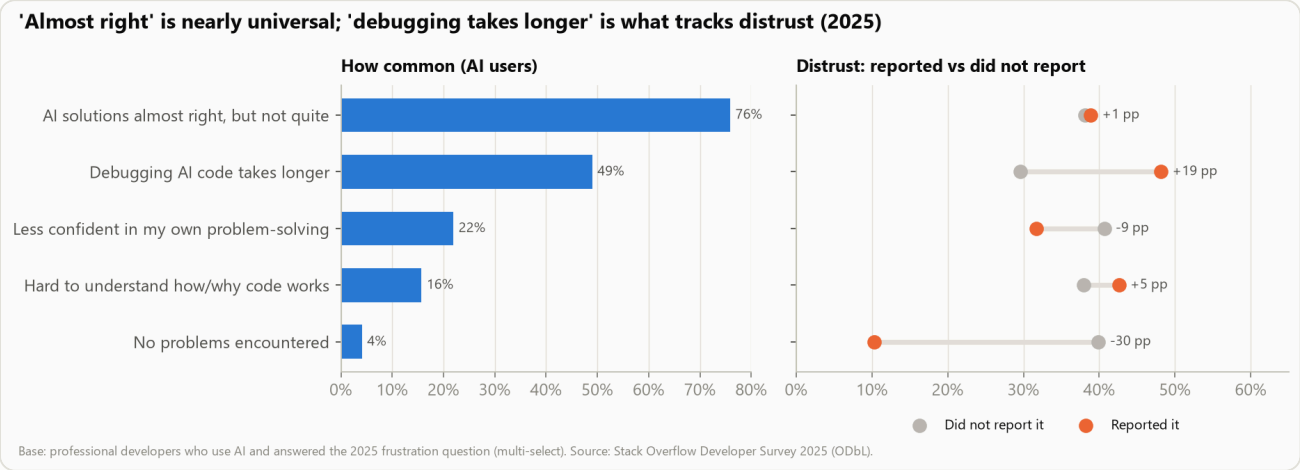
2. Experience is the strongest personal predictor of distrust. In 2025, 25% of AI users with 0–4 years of coding distrust AI, against 43% of those with 30+ years.

- **Size:** with controls, 30+ years vs 0–4 years means about **3× the odds** of distrust.
- **Use patterns explain only part of it.** Adding use frequency, agents, task types and frustrations shrinks the gap from +29 pp to +21 pp. Most of the experience effect remains. Seniors don't distrust AI only because they use it differently; they judge the same output more harshly.
- **Age vs experience:** at a given experience level, *older* developers distrust less. Experience, not age, drives the effect.

3. Use goes with trust: occasional users are the most sceptical. In 2025, distrust is 28% among daily users, 48% among weekly users and 66% among monthly-or-less users.

- **Size:** monthly vs daily means 3.3× the odds, with controls.
- **Agents:** people who use AI agents are 6 pp less likely to distrust than people who don't.
- **Direction is unknown.** Trust may drive use as much as use builds trust.

4. "Almost right" is not the differentiator; "debugging takes longer" is.



2025 frustration (professional AI users)	Share reporting it	Odds ratio for distrust (full model)	Change in P(distrust)
AI solutions "almost right, but not quite"	76%	0.97 (n.s.)	~0 pp
Debugging AI-generated code takes longer	49%	**2.22**	**+15 pp**
Hard to understand how/why the code works	16%	1.14	+2 pp
Less confident in my own problem-solving	22%	0.77	−5 pp
No problems encountered	4%	0.27	−21 pp

- **Every way we tested it,** "almost right" stays at or near zero (odds ratio between 0.89 and 1.08).
- **What matters is what an imperfect answer costs.** Users who say fixing AI code is slower are far more likely to distrust it.

- **An over-reliance warning sign:** users who feel *less* confident in their own problem-solving trust AI *more*.

5. Who is most and least sceptical (2025, AI users).

- **By role:** embedded/hardware developers distrust most (50%), followed by research and DevOps/infra. QA, data/ML and front-end developers distrust least (28–32%).
- **By country:** Switzerland, Sweden and Germany are highest (about 50%), with the UK, US and Canada close behind (47–48%). India (14%) and Ukraine (24%) are lowest, and these gaps persist with controls.
- **By company size:** larger companies look slightly more sceptical in the raw numbers, but there is **no association once role, experience and country are controlled.**
- **By type of AI use:** people who use AI for deployment, predictive analytics, debugging or planning report less distrust. This most likely reflects trusting users delegating more, not those tasks building trust.

What this means

For developer-tools product teams

1. **Optimise for the cost of fixing, not only first-answer accuracy.** Fast verification makes debugging cheaper: inline tests, diffs against the codebase, citations, confidence signals, one-click revert. That cost is the frustration that tracks distrust.
2. **Design for sceptical senior users.** Give them explicit uncertainty, explanations they can check, and controls over scope. Seniors are both the most sceptical group and the people who approve team-wide rollouts.
3. **Move occasional users to habitual use.** The trust gap between monthly and daily users is the largest gap in the data. Onboarding that delivers early, verifiable wins is a trust lever.

For engineering leaders planning AI rollouts

1. **Expect high adoption with low trust, and plan review processes accordingly.** In 2025, 81% of professional developers use AI, but only 38% of users trust its accuracy.
2. **Track debugging and rework time, not just usage.** Rising "debugging takes longer" is an early warning that trust, and the value of the rollout, is eroding.
3. **Watch for over-reliance.** People who lose confidence in their own problem-solving trust AI more, which is a coaching and code-review issue.
4. **Read country and role gaps as context, not destiny.** Pilot with teams whose work matches the tasks where users report the least distrust.

Caveats

- **Self-selected sample.** Respondents are developers engaged enough with Stack Overflow to take a long survey. Results do not represent all developers. Reweighting corrects only the *observed* mix: country, experience, age and company size.
- **2025 missing answers.** The AI section was optional in 2025, and about 31% of 2025 respondents skipped it.
- **Associations, not causes.** The data is cross-sectional and self-reported. For example, trust may cause frequent use rather than the reverse.

- **Changed questions.** Every mapping is documented in `mappings/question_crosswalk.csv`. Questions that exist in only one year (frustrations, agents, use frequency) are analysed within 2025 only.
- **Figures quoted from Stack Overflow** use its official tables, which our pipeline reproduces to within 0.6 pp (84 of 84 checks): 2025 trust 32.8%, distrust 45.7%, "almost right" 66.0%, and 84% using or planning to use AI. Our own trend figures use the comparable "current AI users" base and are labelled as such.

Method: Python (pandas, statsmodels), DuckDB SQL. Logistic regressions with robust standard errors; average marginal effects are reported in percentage points. Robustness checks: ordered logit, "highly distrust" outcome, excluding the US, complete cases, professional experience, reweighted sample, all coders. Full code: `run_pipeline.py`. Data: Stack Overflow Developer Survey 2023–2025, Open Database License (ODbL).